Introduction to Calculations of Heat loss on piping with Solar radiation

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1 Introduction

At some point during one of the project I was a part of, it was necessary to check if the chosen thickness of insulation for long CIP water loops was enough, to keep the temperature control and over certain threshold. It turned out that one of calculations were done in a rush, resulting in wrong decision of making insulation thicker. Just in time, one of senior engineers stepped in and stopped the company from making somewhat pricy mistake.

This senior engineer, later send us the calculaitons to check it, write down things that were wrong and report back to him, to have a conversation about our findings. This way we could learn to spot errors. Because I like calculations and love learning about things inherently related to mathematics it was a match made in heaven. At this point in time I have not started my introductory book to mechanics of heat transfer by great J. Cengel, so it wasn't something I could jump right into. Thus, I took my time, and in around 2 weeks felt somewhat comfortable, at least with concepts (side note: I was especially taken by transient analysis and eigendecomposition of differential equations in plug flow problems, amazing, it might be something to write about) in steady state analysis to begin deriving equations and testing my knowledge.

The original calculations were made using iterative method, which is fine if the iterations are "fairly" small or the change in the underlying function is relatively insignificant or, even better, constant.

Funny note: I'm writting that I felt comfortable in around 2 weeks, but it took me around 5-6 months to understand that my first solution was completely wrong mathematically. Additionally, I had to rewritte my whole code because I convinced myself that all-functions approach is the way to go, which made my code unusable. Then I had to rewritte it again because of abovementioned mathematical error. All in all it took me around 9 months to have working template with this written as well.

2 Basic model and notation

We will be dealing with a lot of variables, and it is important to explain the notation correctly in the beginning. My main basic model assumption is that going from the inside of a piping we have fluid (f),

piping wall (w), insulation layer (i), then surface of insulation (s) and ambient air (a). Because we will be deriving general equations that take into account both external heat flux as well as intergnall heat tracing, we will denote a sliding layer (y) that changes from (i) to (s). It is shown on below diagram.

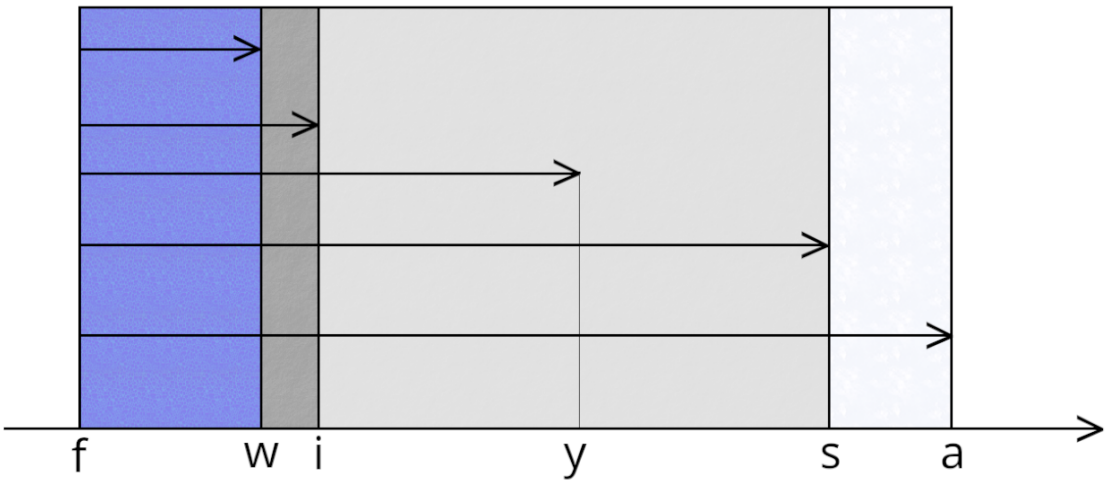


Figure 1: **Piping layering** Shows how layers change with radius. From left: fluid, wall, insulation and air.

Now, let F_{ij} be any variable, then ij index denotes that te variale is measured in relation to layers from i to j. For example, R_{fi} is the combined thermal resistance of fluid and wall layers.

Fluid and surface temperature are a function of length only, even if diameter changes or insulation thickness from one segment to another, those changes are also dependent on length. This is worth keeping in mind, as the dependence will be mostly ommitted.

The value of thermal transfer coeff. around piping surface h_{sa} (h for short) depends on the surface temperature T_s of single piping segment only because of used correlations. This will be always explicilty stated. Of course, as mentioned above surface temperature is a function of length, but in h this will be ommitted for simplicity, and I just think it is easier to think of it as being a function of surface temperature only and not add length dependency.

3 Derivation

3.1 Basics of steady-state heat transfer

The assumptions we must consider to make calculations possible are:

1. Steady-state process.
2. The change in kinetic and potential energy is negligible.
3. The flow is turbulent and pipes are full of liquid.
4. The fluid is incompressible.
5. The specific heat capacity is constant.
6. Fluid is loosing energy to surroundings.
7. No axial heat transfer.
8. Uniform heat transfer distribution in piping cross section.
9. Piping is horizontal and enclosed by surface at ambient temperature separated by air only at ambient pressure and temperature.

10. Internal fluid heat transfer coeff. is insignificant and material conductivities are constant.

11. Absorptivity is equal to unit (heat flux mode).

12. Tracing is at the piping-insulaiton interface (heat tracing mode).

With above in mind, one can utilize Thermal Resistance Network methodology and write that heat transfer in each layer of piping is equal to:

$$\dot{Q}_{ij} = \frac{\Delta T_{ij}}{R_{ij}} \tag{1}$$

The place at which we apply heat flux is different in each calculation mode. In heat tracing we apply heat flux at (y=i) and in external heat flux we apply it at (y=s), because of that the general heat balance can be written as:

$$-\frac{\Delta T_{fy}}{R_{fy}} = \dot{m} C dT_f \tag{2}$$

This will be the form from which we will derive the general equation in the next section. It is also important to define thermal resistance. In general, the total thermal resistance is:

$$R_{fa} = R_{fw} + R_{wi} + R_{is} + R_{sa} = \frac{1}{2\pi L}\left(\frac{2}{h_f D_i} + \frac{1}{k_{pipe}}\ln\left(\frac{D_o}{D_i}\right) + \frac{1}{k_{ins}}\ln\left(\frac{D_t}{D_o}\right) + \frac{2}{hD_t}\right) \tag{3}$$

where:

- L - piping of length L, for which thermal resistance is calculated,
- D_i - inner piping diameter,
- D_o - outer piping diameter,
- D_t - total diameter $D_o + 2 \cdot$ ins.thickness,
- h - combined heat transfer coefficient from surface of a piping to air,
- k_{pipe} - thermal conductivity of a piping,
- k_{ins} - thermal conductivity of a insulation,

Here, because of assumption (10) we let $h_f \to \infty$ in return we have $R_{fw} \to 0$. Thus, form now on we will be writting R_{wi} simply as R_{fi}.

3.2 General solution

3.2.1 Energy balance preparation in finite step temperature change

We will start with analysing finite step analysis of fluid temperature drop over finite length change. Using graph for this would be the simplest approach.

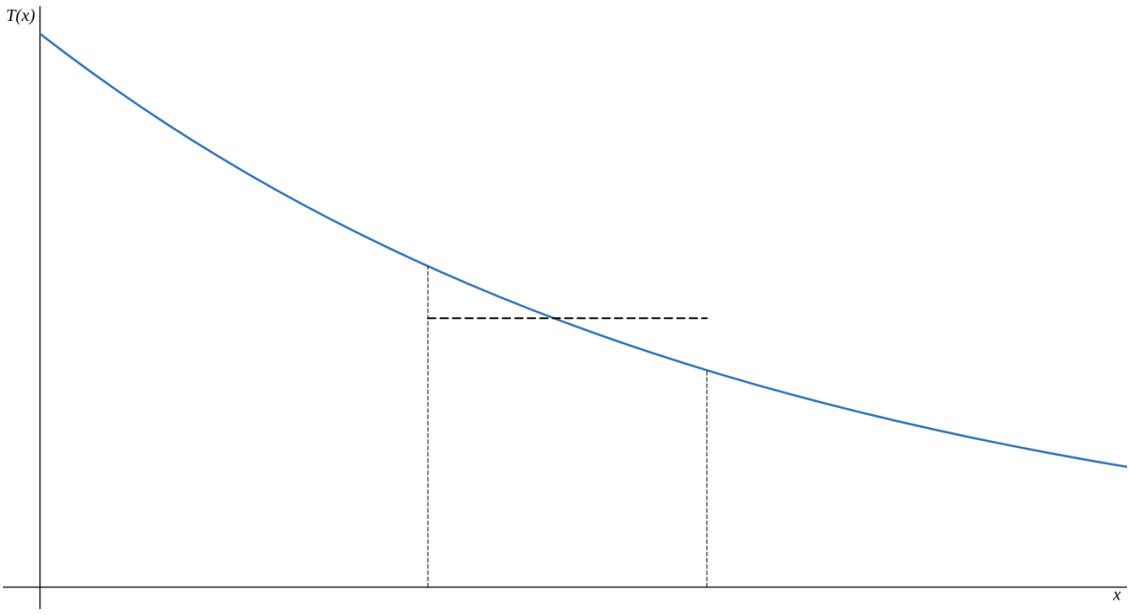


Figure 2: **T(x)** Function of T(x)

In the figure above we consider a change in temperature over a finite change in length on interval $[x_i, x_{i+1}]$, where temperature drops from $T_f(x_i)$ to $T_f(x_{i+1})$. The heat loss over piping segment $x_{i+1} - x_i = \Delta x_i$ is not constant, but drops with temperature. Because of that we consider some average temperature difference to find the average heat loss. The temperature we will use to find the difference is shown on the figure above as a horizontal line between x_i and x_{i+1}. With $w_i \in [0, 1]$ we can write that the average heat loss is:

$$\dot{Q}_{avg}(x_i) = \frac{\Delta T_{fy}}{R_{fy}} = U_{fy} P_t \cdot \Delta T_{fy}(x_i + w_i \Delta x_i) \cdot (x_{i+1} - x_i) = U_{fy} P_t \Delta T_{fy}(x_i^*) \Delta x_i$$

where, for completeness, U_{fy} is the heat transfer coeff. from fluid to sliding layer taken in referecne to total area A_t:

$$U_{fy} = \frac{1}{A_t R_{fy}}, \quad 2\pi L R_{fy} = \frac{1}{k_{pipe}} \ln\left(\frac{D_o}{D_i}\right) + \frac{1}{k_{ins}} \ln\left(\frac{D_y}{D_o}\right)$$

energy balance is:

$$-\dot{Q}_{avg}(x_i) = \dot{m} C \left(T_f(x_{i+1}) - T_f(x_i)\right) = \dot{m} C \Delta T_{f,i}$$

We connect above equations, rearrange it and write:

$$\frac{1}{U_{fy} \Delta T_{fy}(x_i^*)} \frac{\Delta T_{f,i}}{\Delta x_i} = -\frac{P_t}{\dot{m} C} = -\frac{1}{\lambda}$$

We can now take limit at $\Delta x_i \to 0$ and we will arrive at derivative form:

$$\frac{1}{U_{fy} \Delta T_{fy}} \frac{dT_f}{dx} = -\frac{1}{\lambda} \tag{4}$$

And this is basically the equation we must integrate over. Here, the problem is that, we don't know how T_f and T_y are connected to each other and to the applied heat flux. I found that the simplest way is to take cacalculations to the surface temperature domain, as it can be easily connected with all variables. We can then integrate over surface temperature and just translate back to fluid temperature.

3.2.2 Surface temperature domain change

We start by going back to figure (1) and drawing two control systems.

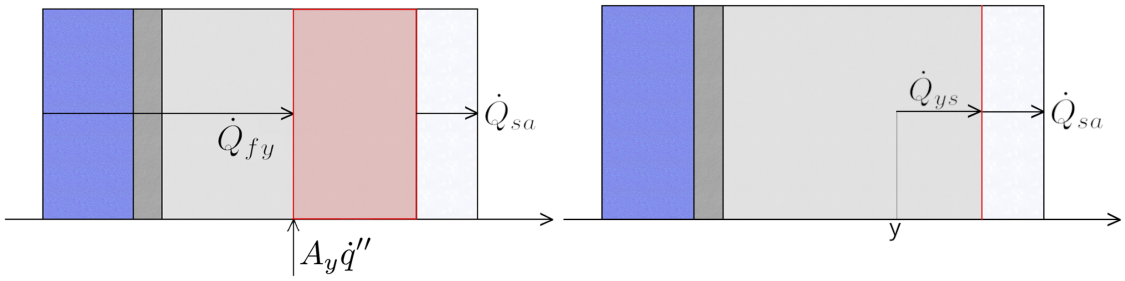


Figure 3: **Control Systems** (1) Control volue at sliding layer; (2) Control boundary at insulation surface

The red lines specify the control system. The first one, a control volume, and the second, a control surface (boundary). With those two, we can write a system of equations fixing the state:

$$A_y\dot{q}'' + \dot{Q}_{fy} = \dot{Q}_{sa}$$
$$\dot{Q}_{ys} = \dot{Q}_{sa}$$

We can substitute eq. (1) for each $\dot{Q}_{ij}$, solve for T_y and substitute bottom equation into the upper one. This allows us to write the general steady state heat transfer equation through piping layers:

$$U_{fy}\Delta T_{fs} + \frac{D_y}{D_t}\dot{q}'' = \left(1 + \frac{U_{fy}}{U_{ys}}\right) h(T_s)\Delta T_{sa} \tag{5}$$

Let us writte the term on the LHS in front of heat transfer coeff. as u_{ry}, this will simplify later calculations. Now, we take differential on both sides getting:

$$U_{fy}\left(dT_f - dT_s\right) = u_{ry}\left(h(T_s)dT_s + \Delta T_{sa}dh_{T_s}\right)$$

$$dT_f = \left(1 + u_{fy}\frac{h(T_s)}{U_{fy}} + u_{ry}\frac{\Delta T_{sa}}{U_{fy}}\frac{dh(T_s)}{dT_s}\right) dT_s \tag{6}$$

One more thing is to specify $U_{fy}\Delta T_{fy}$, which is required by (4). We can take it simply from the first equation in the system of equations presented below fig. (3). And that's basically it. We substitute it and eq. (6) into eq. (4).